

Lab program-5:

5. Construct a scheduling program with C that selects the waiting process with the highest priority to execute next.

```
C lab_5.c > main()
1  #include <stdio.h>
2
3  struct priority_scheduling
4  {
5      char process_name;
6      int burst_time;
7      int waiting_time;
8      int turn_around_time;
9      int priority;
10 };
11
12 int main()
13 {
14     int number_of_process;
15     int total = 0;
16     int ASCII_number = 65;
17     int position;
18
19     float average_waiting_time;
20     float average_turnaround_time;
21
22     printf("Enter the total number of Processes: ");
23     scanf("%d", &number_of_process);
24
25     struct priority_scheduling process[number_of_process];
26     struct priority_scheduling temp_process;
27
28     printf("\nPlease Enter the Burst Time and Priority of each process:\n");
29
30     for (int i = 0; i < number_of_process; i++)
31     {
32         process[i].process_name = (char)ASCII_number;
33
34         printf("\nEnter the details of process %c\n", process[i].process_name);
35
36         printf("Enter the Burst Time: ");
37         scanf("%d", &process[i].burst_time);
38
39         printf("Enter the Priority: ");
40         scanf("%d", &process[i].priority);
41
42         ASCII_number++;
43     }
44
45     // Sort according to priority (Higher number = Higher priority)
46     for (int i = 0; i < number_of_process; i++)
47     {
48         position = i;
49
50         for (int j = i + 1; j < number_of_process; j++)
51         {
52             if (process[j].priority > process[position].priority)
53             {
54                 position = j;
55             }
56         }
57
58         temp_process = process[i];
59         process[i] = process[position];
60         process[position] = temp_process;
61     }
62
63     // Calculate Waiting Time
64     process[0].waiting_time = 0;
65
66     total = 0;
67
68     for (int i = 1; i < number_of_process; i++)
69     {
70         process[i].waiting_time = 0;
71
72         for (int j = 0; j < i; j++)
73         {
74             process[i].waiting_time += process[j].burst_time;
75         }
76
77         total += process[i].waiting_time;
78     }
79
80     average_waiting_time = (float)total / number_of_process;
81
82     // Calculate Turnaround Time
83     total = 0;
84
85     printf("\n\n");
86     printf("-----\n");
87     printf("Process\tBurst Time\tPriority\tWaiting Time\tTurnaround Time\n");
88     printf("-----\n");
89
90     for (int i = 0; i < number_of_process; i++)
91     {
92         process[i].turn_around_time =
93             process[i].burst_time + process[i].waiting_time;
94
95         total += process[i].turn_around_time;
96
97         printf("%c\t%d\t%d\t%d\t%d\n",
98             process[i].process_name,
99             process[i].burst_time,
100            process[i].priority,
101            process[i].waiting_time,
102            process[i].turn_around_time);
103     }
104
105     average_turnaround_time = (float)total / number_of_process;
106
107     printf("-----\n");
108     printf("\nAverage Waiting Time      : %.2f", average_waiting_time);
109     printf("\nAverage Turnaround Time : %.2f\n", average_turnaround_time);
110
111     return 0;
112 }
```

OUTPUT:

```
PS C:\Users\SMILE\Documents\OS\Lab programs> ./lab_5.exe
Please Enter the Burst Time and Priority of each process:

Enter the details of process A
Enter the Burst Time: 4
Enter the Priority: 2

Enter the details of process B
Enter the Burst Time: 3
Enter the Priority: 1

Enter the details of process C
Enter the Burst Time: 5
Enter the Priority: 3

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Process Burst Time      Priority      Waiting Time      Turnaround Time
-----
PC      5              3              0              5
PA      4              2              5              9
PB      3              1              9              12
-----

Average Waiting Time    : 4.67
Average Turnaround Time : 8.67
```